

Q1:- R-D for Inter-arrival Time.

| I-A Time | Probability | Cumulative Prob |
|----------|-------------|-----------------|
| 1        | 0.125       | 0.125           |
| 2        | 0.125       | 0.250           |
| 3        | 0.125       | 0.375           |
| 4        | 0.125       | 0.500           |
| 5        | 0.125       | 0.625           |
| 6        | 0.125       | 0.750           |
| 7        | 0.125       | 0.875           |
| 8        | 0.125       | 1.000           |

R-D Assignment-

|            |
|------------|
| 001 - 125  |
| 126 - 250  |
| 251 - 375  |
| 376 - 500  |
| 501 - 625  |
| 626 - 750  |
| 751 - 875  |
| 876 - 1000 |

R-D for Service Time.

| Service Time | Prob | Cumulative Prob |
|--------------|------|-----------------|
| 1            | 0.17 | 0.17            |
| 2            | 0.15 | 0.32            |
| 3            | 0.32 | 0.64            |
| 4            | 0.20 | 0.86            |
| 5            | 0.06 | 0.90            |
| 6            | 0.10 | 1.00            |

R-D Assignment-

|        |
|--------|
| 01-17  |
| 18-32  |
| 33-64  |
| 65-86  |
| 87-90  |
| 91-100 |

Simulation Table

| Customer | R-D Arrival | I-A Time | CLK Time of Arrival | R-D Service Time | Service Time | S-T Bginn | Time customer waits in phone | S-T Ends | Time spent in system | Total time of service |
|----------|-------------|----------|---------------------|------------------|--------------|-----------|------------------------------|----------|----------------------|-----------------------|
| 1        | -           | 1        | 0                   | 74               | 4            | 0         | 0                            | 4        | 4                    | 0                     |
| 2        | 751         | 7        | 7                   | 52               | 3            | 7         | 0                            | 10       | 3                    | 3                     |
| 3        | 303         | 3        | 10                  | 16               | 1            | 10        | 0                            | 11       | 1                    | 0                     |
| 4        | 106         | 1        | 11                  | 82               | 4            | 11        | 0                            | 15       | 4                    | 0                     |
| 5        | 94          | 1        | 12                  | 94               | 6            | 15        | 3                            | 21       | 9                    | 0                     |
| 6        | 606         | 5        | 17                  | 61               | 3            | 21        | 4                            | 24       | 7                    | 8                     |
| 7        | 747         | 6        | 23                  | 87               | 5            | 24        | 1                            | 29       | 6                    | 0                     |
| 8        | 339         | 3        | 26                  | 35               | 3            | 29        | 3                            | 32       | 6                    | 0                     |
| 9        | 877         | 8        | 34                  | 29               | 2            | 34        | 0                            | 36       | 2                    | 2                     |
| 10       | 454         | 4        | 38                  | 99               | 6            | 38        | 0                            | 44       | 6                    | 2                     |

→ Server utilization =  $37/44 = 0.771$   
 → Avg waiting time =  $11/10 = 1.1$   
 → Avg service time =  $37/10 = 3.7$

→ Avg time spent by customer in the system =  $48/10 = 4.8$



Q8:-

| <u>No of customers</u> | <u>Prob</u> | <u>CDF</u> | <u>Random disit</u>  |
|------------------------|-------------|------------|----------------------|
| 4                      | 0.25        | 0.25       | 0 - <del>24</del> 25 |
| 5                      | 0.33        | 0.58       | 25 - 58              |
| 6                      | 0.27        | 0.85       | 59 - 85              |
| 7                      | 0.15        | 1.00       | 86 - 99.             |

### order distribution

| <u>No of dozens</u> | <u>Prob</u> | <u>CDF</u> | <u>Random disit</u> |
|---------------------|-------------|------------|---------------------|
| 1                   | 0.20        | 0.20       | 0 - 20              |
| 2                   | 0.35        | 0.55       | 21 - 55             |
| 3                   | 0.15        | 0.70       | 56 - 70             |
| 4                   | 0.30        | 1.00       | 71 - 99.            |

\* 18 dozen baked everyday  $\rightarrow$  cost Price  $18 \times 3 = \boxed{54}$

| <u>Days</u> | <u>No of customer</u>   | <u>cust</u>                     | <u>Dozen ordered</u>                         | <u>Revenue generated</u>                           | <u>Loss</u> | <u>Salvage Po.</u>        | <u>Daily Profit</u>                              |
|-------------|-------------------------|---------------------------------|----------------------------------------------|----------------------------------------------------|-------------|---------------------------|--------------------------------------------------|
| 1           | 5<br>(from<br>(34 R.D)) | 1<br>2<br>3<br>4<br>5           | 1<br>2<br>2<br>4<br>3 }<br>↓ from R.D.<br>12 | Total<br>dozen<br>sold.<br>$12 \times 5$<br>$= 60$ | 0           | $6 \times 2.5$<br>$= 15$  | $60 - 54 = 0$<br>$+ 15$<br>$= \underline{21}$    |
| 2           | 7<br>(from              | 1<br>2<br>3<br>4<br>5<br>6<br>7 | 1<br>2<br>4<br>2<br>4<br>3<br>1 }<br>17      | $17 \times 5$<br>$= 85$                            | 0           | $1 \times 2.5$<br>$= 2.5$ | $85 - 54 = 0$<br>$+ 2.5$<br>$= \underline{33.5}$ |
| ∴           |                         | <u>Total Profit</u>             |                                              | $21 + 33.5 = \underline{54.5 \text{ Rs}}$          |             |                           |                                                  |



# Q2:- Event Scheduling / Time Advance Algorithm

Event Scheduling Algo Explain  $\rightarrow$  4 Marks.

Event Scheduling Table  $\rightarrow$  6 M.

| <u>clock</u> | <u>System</u><br><u>Left</u> | <u>state</u><br><u>Left</u> | <u>Future Event-list</u> | <u>Cumulative</u><br><u>Statistics</u><br><u>B</u> | <u>MPL</u> |
|--------------|------------------------------|-----------------------------|--------------------------|----------------------------------------------------|------------|
|              |                              |                             | (D, 2) (A, 3) (E, 20)    | 0                                                  | 0          |
| 0            | 0                            | 1                           | (A, 3) (E, 20)           | 2                                                  | 0          |
| 2            | 0                            | 0                           | (A, 5) (D, 8) (E, 20)    | 2                                                  | 0          |
| 3            | 0                            | 1                           | (D, 8) (A, 11) (E, 20)   | 7                                                  | 1          |
| 5            | 1                            | 1                           | (A, 11) (D, 13) (E, 20)  | 7                                                  | 1          |
| 8            | 0                            | 0                           | (A, 15) (D, 19) (E, 20)  | 7                                                  | 1          |
| 11           | 0                            | 1                           | (A, 17) (D, 19) (E, 20)  | 9                                                  | 2          |
| 13           | 1                            | 1                           | (D, 19) (A, 21) (E, 20)  | 13                                                 | 2          |
| 17           | 0                            | 1                           | (A, 21) (A, 24) (E, 20)  | 15                                                 | 2          |
| 19           | 1                            | 1                           |                          |                                                    |            |

Q3 -

1) For a Poisson distribution

→ Mean  $\mu = \lambda$ .

→ Variance  $\sigma^2 = \lambda$ .

→ Std. deviation  $\sigma = \sqrt{\lambda}$ .

$$\sigma = \sqrt{6} = \underline{2.449}$$

→ Threshold = mean + 1 std deviation.

$$k = \mu + \sigma \approx 6 + 2.449 = 8.449 \approx \underline{9}$$

$$\therefore P(X > 8) = 1 - P(X \leq 8)$$

→ Cumulative Poisson Probability

$$P(X \leq 8) = \sum_{x=0}^8 \frac{e^{-\lambda} \lambda^x}{x!}$$

$$\lambda = 6$$

$$P(X \leq 8) \approx 0.8472$$

$$\therefore P(X > 8) = 1 - 0.8472 = \underline{0.1528}$$

$\therefore$  Roughly 15% chance that Tom will receive more than 1 std. deviation above the average (i.e. 9 or more emails) tomorrow.



## 2) EXPONENTIAL distribution

→ For an exponentially random variable  $T$  with mean  $\mu = 60$

$$f_T(t) = \frac{1}{\mu} e^{-t/\mu}, \quad t \geq 0.$$

$$F_T(t) = P(T \leq t) = 1 - e^{-t/\mu}$$

→ Probability for a single customer

$$\begin{aligned} P(T < 60) &= 1 - e^{-60/60} = 1 - e^{-1} \\ &= 1 - 0.367879 \approx \underline{0.6321} \end{aligned}$$

So each customer has 63.21% chance to finish in less than 60 seconds.

→  $\therefore$  Probability for two customers (independent).

Service times are independent.

$$\therefore P(\text{both customers} < 60 \text{ Sec}) = P(T_1 < 60) \times P(T_2 < 60).$$

$$= 0.6321 \times 0.6321$$

$$= \underline{0.3996}$$

$\therefore$  Roughly 40% chance that both customers in front will finish their transactions in under 60 seconds each.

Q3 (iii). M/M/1 Queue

→ Arrival rate  $\lambda = 45$  per hr.

→ Service rate  $\mu = 1$  per minute = 60 per hr.

→ single server → M/M/1 Queue.

$$\rho = \lambda / \mu = 45 / 60 = \underline{0.75}$$

Steady state Parameters

$$1) L = \frac{\rho}{1 - \rho} = \frac{0.75}{1 - 0.75} = \underline{3}$$

$$2) L_q = \frac{\rho^2}{1 - \rho} = \frac{(0.75)^2}{1 - 0.75} = \underline{2.25}$$

$$3) W = \frac{1}{\mu - \lambda} = \frac{1}{60 - 45} \text{ hr} \\ = \frac{1}{15} \text{ hrs} = \underline{4 \text{ minutes}}$$

$$4) W_q = \frac{\rho}{\mu - \lambda} = \frac{0.75}{60 - 45} = \frac{0.75}{15} \\ = 0.05 \text{ hr} = \underline{3 \text{ minutes}}$$